1) $\lim\limits_{x\to 0}\dfrac{\sqrt{1+x}-1}{x}=\left(\dfrac{0}{0}\right)=\lim\limits_{x\to 0}\dfrac{\left(\sqrt{1+x}-1\right)\left(\sqrt{1+x}+1\right)}{x\left(\sqrt{1+x}+1\right)}=\lim\limits_{x\to 0}\dfrac{(1+x-1)}{x\left(\sqrt{1+x}+1\right)}=$

$$=\lim_{x\to 0}\frac{x}{x\left(\sqrt{1+x}+1\right)}=\lim_{x\to 0}\frac{1}{\left(\sqrt{1+x}+1\right)}=0.5$$

$$0x=0$$
$$0x=4$$

2) $\lim\limits_{x\to 1}\dfrac{\sqrt[3]{2x-1}-\sqrt[3]{3x-2}}{x-1}=\left(\dfrac{0}{0}\right)=\lim\limits_{x\to 1}\dfrac{\left(\sqrt[3]{2x-1}-\sqrt[3]{3x-2}\right)\left(\sqrt[3]{(2x-1)^2}+\sqrt[3]{2x-1}\sqrt[3]{3x-2}+\left(\sqrt[3]{3x-2}\right)^2\right)}{(x-1)\left(\sqrt[3]{(2x-1)^2}+\sqrt[3]{2x-1}\sqrt[3]{3x-2}+\left(\sqrt[3]{3x-2}\right)^2\right)}=$

$$=\lim_{x\to 1}\frac{(2x-1)-(3x-2)}{(x-1)\left(\sqrt[3]{(2x-1)^2}+\sqrt[3]{2x-1}\sqrt[3]{3x-2}+\left(\sqrt[3]{3x-2}\right)^2\right)}=$$
$$=\lim_{x\to 1}\frac{-(x-1)}{(x-1)\left(\sqrt[3]{(2x-1)^2}+\sqrt[3]{2x-1}\sqrt[3]{3x-2}+\left(\sqrt[3]{3x-2}\right)^2\right)}=$$
$$=\lim_{x\to 1}\frac{-1}{\left(\sqrt[3]{(2x-1)^2}+\sqrt[3]{2x-1}\sqrt[3]{3x-2}+\left(\sqrt[3]{3x-2}\right)^2\right)}=-\frac{1}{3}$$

$$a^3\pm b^3=(a\pm b)\left(a^2\mp ab+b^2\right)$$
$$a^7-b^7=(a-b)\left(a^6+a^5b+a^4b^2+a^3b^3+a^2b^4+ab^5+b^6\right)$$
$$a^7+b^7=(a+b)\left(a^6-a^5b+a^4b^2-a^3b^3+a^2b^4-ab^5+b^6\right)$$

3) $\lim\limits_{x\to 0^+}(x\cdot\ln x)=(0\cdot(-\infty))=\lim\limits_{x\to 0^+}\dfrac{(\ln x)}{\frac{1}{x}}=\begin{pmatrix}t=\frac{1}{x}\\ x=\frac{1}{t}\\ t\to+\infty\end{pmatrix}=\lim\limits_{t\to+\infty}\dfrac{\left(\ln\left(\frac{1}{t}\right)\right)}{t}=\lim\limits_{t\to+\infty}\dfrac{\ln\left(t^{-1}\right)}{t}=$

$$=\lim_{t\to+\infty}\frac{-1\cdot\ln t}{t}=0$$

$$\frac{x\cdot\ln x}{1}=\frac{\ln x}{\frac{1}{x}}$$

4) $\lim\limits_{x\to 0}\dfrac{e^{x^2}-\cos x}{\left(\sin\ x\right)^2}=\left\{\dfrac{0}{0}\right\}=\lim\limits_{x\to 0}\dfrac{(e^{x^2}-1)+(1-\cos x)}{x^2}=$

$$=\lim_{x\to 0}\frac{(e^{x^2}-1)}{x^2}+\lim_{x\to 0}\frac{(1-\cos x)}{x^2}=\lim_{x\to 0}\frac{(e^{x^2}-1)}{x^2}+\frac{1}{2}=\begin{pmatrix}t=x^2\\ t\to 0\end{pmatrix}$$
$$=\lim_{t\to 0}\frac{(e^t-1)}{t}+\frac{1}{2}=1+\frac{1}{2}=1.5$$

Первый замечательный предел $\lim\limits_{x\to 0}\dfrac{\sin x}{x}=1$

$$\lim_{x\to 0}\frac{x}{1-e^x}=\lim_{x\to 0}-\frac{x}{\left(e^x-1\right)}=-1$$
$$\lim_{x\to 1}\frac{x-1}{x^2-1}=\frac{1}{2}$$

$$\lim_{x\to 0}\frac{x}{x^2}=\infty$$
$$\lim_{x\to 0}\frac{sinx}{x^2}=\lim_{x\to 0}\frac{sinx}{x}\cdot\frac{1}{x}=\infty$$

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